Some Problems in Valuation

Week 9: Mergers, MBOs & other Corporate Reorganizations

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Outline

1. As interest rates change, how does the cost of equity change?
2. Are government securities the right benchmark for the risk free rate (i.e. zero beta) in the CAPM?
3. How can we use the debt markets to improve our estimates of the cost of capital?
4. Project valuation with high leverage.

Appendix 1: An example of using cost of debt as a check on cost of capital

Appendix 2: Calculating a debt beta

Appendix 3: WACC versus the Adjusted Present Value Approach when leverage is changing over the life of the project.

1. How does cost of equity change when the risk free rate changes? Traditional approach

- The expected return on equity using traditional CAPM:

$$E(R_e) = R_f + \beta_e \, (E(R_m) - R_f)$$

We usually use a historic market equity risk premium (MERP) for ($E(R_m) - R_f$) like 5-6%

If R_f declines by say 2%, then how much do we expect E(*Rm*) to decline by? If you say 2 percent, then you are assuming the Market Equity Risk Premium is not changing.

Since 2007 we have seen large changes in both nominal and real interest rates sometimes with swings of up to 4%. So this is an important question.

Two strongly competing views of the market equity risk premium.

1. It is constant (Dimson, Marsh and Staunton!

(Credit Suisse Global Investment Returns Yearbook 2017, p. 41)

"There are good reasons to expect the [market] equity premium to vary over time. Market volatility clearly fluctuates, and investors' risk aversion also varies over time. However, these effects are likely to be brief … For practical purposes, we conclude that to forecast long-run equity premium, it is hard to beat extrapolation from the longest history available …"

A second view: market equity risk premium is not constant and will vary with changes in economic uncertainty

- Bank of England model predicts that consumers and investors will respond to an increase (decrease) in economic uncertainty by increasing (decreasing) demand for risk free assets and reducing (increasing) demand for risky assets. Higher economic uncertainty simultaneously puts downward pressure on the Rf rate and upward pressure on the market equity risk premium.
- When the distribution of expected consumption and GDP growth is more negatively skewed and has a higher probability of extreme events (kurtosis), the market equity risk premium is higher and the Rf rate is lower.
- Evidence that the distribution of GDP growth has become more negatively skewed and tails have become fatter i.e. Kurtosis. The latter reflects the probability of extreme events (the equivalent of financial earthquakes).
- Extreme assumption: Total Market Returns (TMR) is constant and changes in Rf offset by changes in MERP; where TMR= Rf+ MERP

How does the media refer to this increase in economic uncertainty? Referred to as a 'Flight to safety (FTS)?

- Flight to safety refers to sudden shift in investment behaviour in periods of financial turmoil when investors sell high risk investments and purchase safe investments.
- Episodes include Black Monday, Russian debt default, failure of Long Term Capital Management, subprime mortgage crisis in 2008 & Coronavirus. Often accompanied by a deterioration in liquidity.
- "Government bonds kicked off the year with a sharp rally with investors darting into 'havens' amid a global flight from risk sparked by disappointing data on China's sprawling manufacturing sector." FT JAN 2, 2019
- The yield on…ten year US Treasuries bore down on a record low as the spread of coronavirus prompted another flight to safety of government debt, … yesterday's bond market rally was accompanied by a big equity market sell-off. (FTSE fell 3.3%) FT Feb 25, 2020

Flight to safety episodes results in increase in *negative* correlation between returns on stocks and returns on bonds

"Over our sample period, which starts fairly late in 1980, this correlation [between stock and bond returns] is negative for 23 countries. It is conceivable that this negative correlation is mainly caused by the relatively high incidence of flight to safety in the last 30 years" … they go on further to say after analysing Flight To Safety (FTS) periods that "The stock bond correlation in normal periods is -2.1% on average … and The average correlation in FTS times is in fact -42.86% ... Thus, FTS events indeed render the bond equity return correlation substantially more negative."

Page 20, Flight to Safety, Baele, Bekaert, & Inghelbrecht, 2014.

The implications

- If you believe the traditional CAPM view then a decline in the Rf rate will reduce the cost of equity capital.
- If you believe the evidence on ‘flight to safety’ then a decline in the risk free rate will be (partially) offset by a rise in the Equity Risk Premium.
- In an extreme interpretation the decline (increase) in the rf will be offset by a rise (fall) in the MERP. Thus, the Total Market Return is assumed constant.

2. What is the right risk free asset to use in the CAPM?

- Is it the return on a government security such as treasury bills or bonds? In CAPM, a risk free asset is a zero beta asset. Are government securities zero beta assets?
- Government securities are 'special' and investors hold them for reasons other than their expected return. Some institutions are obliged to hold government bonds: banks & insurance companies.
- Concept of 'specialness', gives rise to a convenience yield. This increases the demand for government bonds, raises their price and lowers their yield. Implies that their yield is lower than that on a zero beta asset.
- Then what is a good proxy for a zero beta asset? Possible candidates: yields on AAA or AA corporate bonds.

Some academic views

- Berk and DeMarzo: ‘practitioners sometimes use rates from the highest quality corporate bonds in place of the Treasury rates [in the CAPM] equation.’
- Krishnamarthy & Vissing-Jorgensen (2012, JPE): treasury interest rates are not an appropriate benchmark for `riskless’ rates. **Cost of capital computations using the capital asset pricing model should use a higher riskless rate than the Treasury rate**; a company with a beta of zero cannot raise funds at the Treasury rate.
- Feldhutter & Lando (2008, JFE): **The premium is a convenience yield on holding Treasury securities arising from**, because for example, (a) Treasuries are an important instrument for hedging interest rate risk, (b) Treasury securities must be purchased by financial institutions to fulfil regulatory requirements, (c) that the amount of capital required to be held by a bank is significantly smaller to support an investment in Treasury securities relative to other securities because it has negligible default risk.
- What is the size of the ‘convenience yield’? Varies over time for example with Flight to Safety. Berk and DeMarzo suggest taking the spread between AAA rated bonds and Government bonds. Feldhutter and Lando give a range of 30-90 basis points.

3. Debt markets as a check on our cost of capital estimation?

- For the same company debt claims are less risky than equity claims, because debt claims rank senior to equity claims
- Thus, expected returns on equity are above expected returns on debt.
- Then beta of equity must be above beta of debt
- It also follows that asset beta must be above debt beta because asset beta is a linear combination of equity beta and debt beta.
- $\beta_a = \beta_e$ E/V + β_d D/V
- Thus, risk premium on assets is greater than the risk premium on debt for the same company.

Defining risk premium on assets and risk premium on debt

- E(Ra) = Rf + β_a (MERP) where β_a is asset beta
- E(Rd) = Yield to maturity
- Since Rf rate is the same for both returns on asset (i.e. unlevered cost of equity) and cost of debt, we can focus on the risk premium of each:
  - The Asset Risk Premium (ARP) = β_a (MERP)
  - The Debt Risk Premium (DRP) = YTM – Rf

ARP> DRP

What should be the size of differential (ARP – DRP)?

*For Debt Risk Premium we should adjust for expected loss rate in default

UK regulator publishes (ARP-DRP) differential for a utility company.

Market Equity risk premium	4.77% (Rf =3.86%)
Debt beta	0.075
Asset beta	0.349
Equity beta	0.760
Gearing (leverage)	60%
Asset Risk Premium (ARP)	1.66%
Yield to Maturity (YTM)	5.62%
Expected loss	0.3
Debt Risk Premium (DRP)	1.46%
ARP – DRP differential	0.20%

Is 0.2% a sufficiently large premium?
Debt risk premium = YTM – Rf – expected loss
Suppose you omitted the Expected loss, ARP- DRP differential would go negative

How can we get an estimate of (ARP-DRP) differential for a particular company?

- Use data for the company if its equity & bonds are traded
- Use data from a sample of comparable companies
- Sample should have both equity and bonds traded
- Should be comparable in terms of asset beta and leverage
- Estimate (ARP-DRP) differential for the sample.
- Compute (ARP-DRP) over time. See how variable it is.

4. Calculating discount rates & PV for project financing transactions e.g. PE, infrastructure

- In a PE or infrastructure transaction the debt to value ratio depends upon the (present) value of the project's cash flows. That debt to value ratio is changing over time since free cash flow is often used to repay debt. The result is that if you use the WACC you must use different discount rates for different periods.

- If you know market value or present value of the assets every period then it is easy to calculate the market value leverage ratio and the discount rate.

- But suppose you don't know the value of the assets, because that is what you are trying to find out. Then it is more complex to calculate the leverage ratio and the discount rate.
- In this case you must solve for the leverage ratio and the value of the assets simultaneously.

- Is there a simple approach to this problem?

Valuation when the debt ratio depends upon the value of the project

Value a PE firm with the following constant expected annual CFs in perpetuity:

EBIT	600
Interest payments	60
Taxable income	540
Taxes (20%)	108
After tax income	432

The asset beta is 1.2. The risk free rate is 7%, the expected return on (high yield) debt is 10%. Assume the debt beta is 0.5, and the risk premium on the market is 6%. The tax rate is 20%.

The firm has debt of £600 outstanding in perpetuity. The book value of the assets is currently £3000.

Which FCF do we use in valuing the enterprise value with a WACC or APV calculation? Is it

432, 492 or 480?

(i) Calculate the Adjusted Present Value (APV) of the firm.

APV = Value of all equity financed firm + PV of Financing Benefit

Step 1: calculate the all equity i.e. the unlevered discount rate

R_a = 7% + 1.2 (6%) = 14.2%

PV of all equity financed firm:

All equity cash flows 600 (1-.2) = 480

PV = 480/0.142 = **3380.28**

Step 2: calculate PV of tax shield on leverage = (60 x .2)/0.142 = **84.51**

PV (firm) = 3380.28 + 84.51 = **3464.79**

1. The all equity cash flows are cash flows before interest but after taxes. The tax calculation omits the interest tax shields since these are separately valued in Step 2. Don't double count!
2. Tax shields are assumed as risky as the operating cash flows. This is assumption of our WACC formula.

(ii) If we c**alculate a WACC we must know the market leverage ratio, We can calculate it by iteration.**

<u>Textbook WACC</u>

What is the market value gearing ratio. Problem: you do not know the value of the firm i.e. the enterprise value. You could take the easy way out and use the gearing ratio (600/3464.79) from the APV answer. But you may have to show the WACC calculations. We do it by iteration. First, our initial estimate of the gearing ratio is based on the book value of the assets i.e. D/V = 600/3000 = 0.2

WACC = i (1-T) D/V + R_e E/V

WACC = .1(1-.2) .2 + R_e x .8

Where R_e = 7% + 1.375 (6%)

= 15.25%

β_a = β_e E/V + β_d D/V

1.2 = β_e x 0.8 + 0.5 x 0.2

β_e = 1.375

WACC = .1(1-.2) .2 + .1525 x .8

= 13.8%

PV = 480/.138 = **3478.26**

This is not our final value of the project because the gearing ratio based on our new estimate of the value of the firm: 600/3478.26 = .1725 is not the same as our initial one of 20%. Our initial debt ration of 20% must equal our final debt ratio. We have not converged.

Step 1: Recalculate a new β_e based on leverage estimate of 17.25%

Use formula for beta asset: β_a = β_e E/V + β_d D/V
1.2 = β_e x 0.8275 + (.5 x .1725)
β_e = 1.3459

Step 2: Recalculate a new R_e = 7% + 1.3459 (6%) = 15.076%

Step 3: Calculate WACC = .10(1-.2) .1725 + .15076 x .8275 = .13855 or 13.855%

PV = 480/ .13855 = **3464.45**

We could continue iterating using a new gearing ratio of 600/3464.45 or 17.3%. We can see this is close to our starting debt ration of 17.25%. Or we can just compare it with the APV value.

Notice that could have used the gearing ratio from the APV (600/3464.79) = .1731 and then calculate the WACC.

An important assumption in these WACC calculations: the firm's tax shields are risky

- APV & WACC gave the same answers because we assumed that the tax shields were risky and were discounted at R_a (and not at R_d). Risky because leverage ratio is constant and therefore dollar leverage varies with profitability and value. This is the assumption imbedded in the particular WACC formula we have used. Thus, if value of firm doubles (halves) the $ leverage must double (halve) to keep the leverage ratio constant.
- But suppose debt is fixed & constant in dollars. The tax shields are then less risky than the asset's cash flows & we should use a lower discount rate. What rate,? say R_d. then the two methods will not give the same answer.

What should you do?

If leverage ratio is not constant and debt is fixed in $ terms textbook WACC is not correct: 2 solutions

- Use APV and discount tax shields at R_d.

- Or, use another WACC method: MM WACC

  WACC (mm)= r (1$-T_c$D/V) where r is return on assets

- But here is a catch. For MM WACC there is a different formula for levering/unlevering beta:

  β_e = β_a + (1-Tc)(β_a $-$ β_d)D/E

See Brealey Myers and Allen, 13 th edition: pages 522-523

Appendix 1: Using debt markets for a check on cost of capital: an example

- WACC= 4.664%
- Cost of equity E(Re)= 6.625%
- Cost of debt E(Rd) = 5.62% (Yield to maturity)
- Risk free rate = 4%
- D/V = 60%

- Market equity risk premium = 4.77%
- Beta equity = 0.760; Beta of debt=0.075*; D/V=60%

* You need this if you want to calculate asset beta

Here is one way

- Asset beta = 0.349 (obtained by unlevered equity beta)

Asset risk premium 0.349 * 4.77% (MERP) = 1.66%

Debt risk premium = 5.62% (YTM) – 4% (Rf) = 1.62%

Appendix 2: Calculating the debt beta

- Equation for asset beta (where D= debt, E=equity, and Convertible)

$$\overline{\beta_A} = \frac{D}{V}\beta_D + \frac{E}{V}\beta_E + \frac{C}{V}\beta_C$$

- Must include expected return on debt in WACC equation

  WACC = [E(R_d)D/V + E(R_e) E/V].

Frequently debt beta is assumed to be zero. Imagine a highly levered deal then assuming β_d = 0 will under estimate asset beta, for a given equity beta.

Suppose Beta debt is actually 0.2, D/V =0.5 and Beta equity is 0.8. What is implied Beta asset? 0.5!

Suppose beta debt assumed zero implied Beta asset is 0.4

What is the effect on WACC calculation? You understate WACC.

What assumptions do UK regulators make?

- Many UK regulators assume debt beta is not zero.

- Ofcom (telecoms regulator) use to assume a non zero debt beta when the debt premium is > 100 b.p. It has imposed a debt beta of 0.16.

- Competition Markets Authority assumes a debt beta of 0.075.

But this is for companies with low asset betas & investment grade debt.

- How to calculate debt betas?

Methods for calculating a debt beta

- 2 main methods used by practitioners:

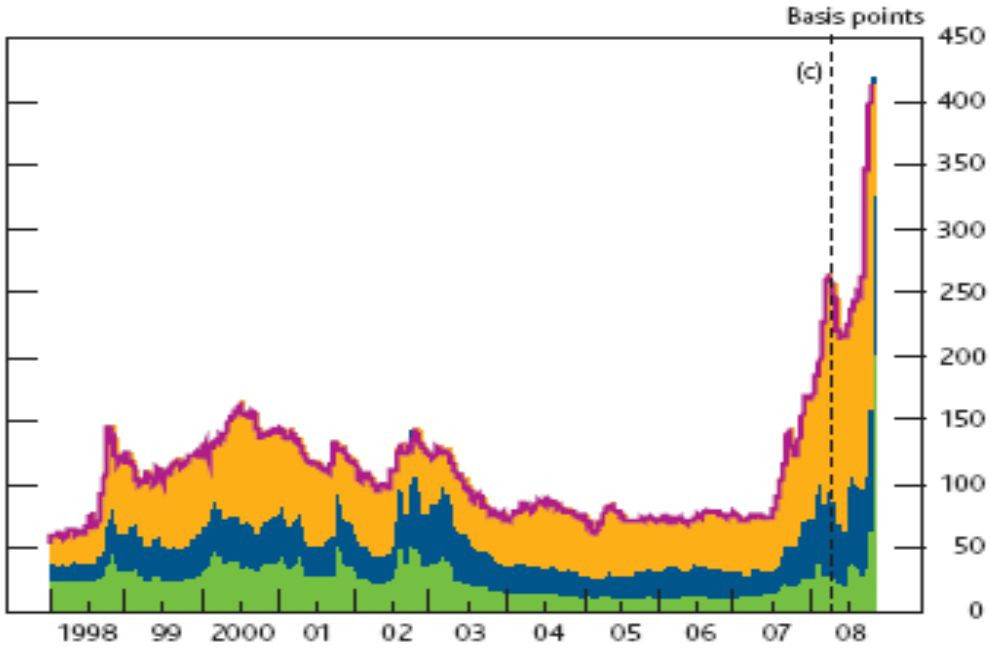
    1. Regression method: regressing debt returns against an equity market index; (there are more complex regressions)
    2. Credit spread decomposition approach

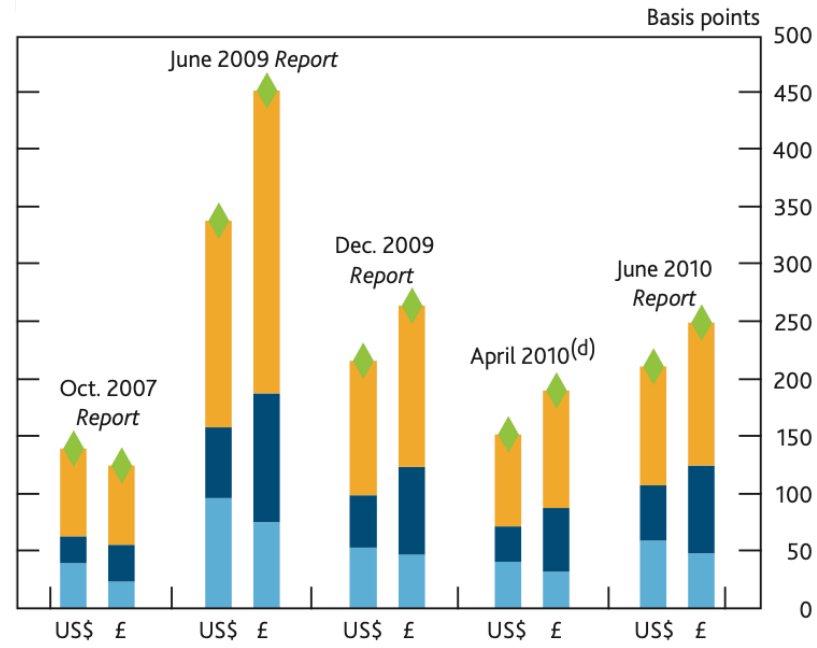
Decomposition of debt premium

- Debt premium (or yield spread): excess of promised yield above equivalent risk free rate.

- Components of debt premium:
  - Default premium: compensation for probability of default and loss rate in default (cash flow adjustment)
  - Premium for non-systematic factors such as liquidity premium
  - Systematic risk premium (input into debt beta): so if the spread related to systematic is 50 bp, and equity risk premium is 500 bp, then debt beta is 0.1.

Sterling-denominated investment-grade corporate bond spreads





- Residual (including compensation for illiquidity)
- Compensation for uncertainty about default losses
- Compensation for expected default losses
- Total

- Actual
- Residual (including compensation for illiquidity)
- Compensation for uncertainty about default losses
- Compensation for expected default losses

Sources: Bank of America Merrill Lynch, Bloomberg, Thomson Reuters Datastream and Bank calculations.

(a) Webber, L and Churm, R (2007), 'Decomposing corporate bond spreads', *Bank of England Quarterly Bulletin*, Vol. 47, No. 4, pages 533–41.
(b) Option-adjusted spreads over government bond yields.
(c) Data to close of business on 14 June 2010.
(d) Trough in spreads on 23 April 2010.

Adjusting the promised return for credit risk in the WACC equation

- Adjust the promised yield for the probability of default and the expected loss rate (call this the default premium).
- Cooper & Davydenko (2007) suggest that for bonds with a debt premium of 100 b.p. the expected default loss is 16 b.p. and the returns premium is 84 b.p so most of the promised yield is an expected return premium. So 16 bp comes off yield to maturity
- In contrast for a bond with a debt premium of 400 b.p. less than one half of the debt spread is an expected returns premium. The majority of the debt premium (about 60%) is caused by the risk of default. So take off 240 bp off yield to maturity.

Appendix 3: A case study where leverage is changing over time

- Medi Media
- Publisher firm acquired in a leveraged buyout.
- Book leverage declining with time according to repayment schedule.
- Question: what value should be placed on the buyout?

Summary

- WACC = $66.5 millions using book leverage to calculate WACC each year

- APV = $57.27 millions

- WACC = $57.31 millions using an annual WACC based on present values
- Error in using book leverage is about $9 millions or about 16% of APV. Do you care? Yes if you are buying or selling a business.

A crude approximation: Yearly WACC based on that year's book value of leverage: Value of equity (E) = $66.5 ml

	A	B	C	D	E	F	G	H	I	J
22	**WACC Method Using Miles-Ezzel**	At Closing	1991	1992	1993	1994	1995	1996	1997	1998
23										
24	**Estimate of Annual Discount Factors**									
25	Average D/E		4.16	2.92	2.08	1.46	1.03	0.71	0.48	0.28
26	Book D/V		**81%**	74%	67%	59%	51%	41%	33%	22%
27										
28	Cost of Debt	9.2%								
29	Opportunity cost of Capital (r_a)	13.61%								
30	Tax Rate	35%								
31	Adjusted Cost of Capital		10.92%	11.13%	11.36%	11.63%	11.92%	12.23%	12.52%	12.88%
32	Annual Discount Factor		0.90	0.81	0.73	0.65	0.58	0.52	0.46	0.41
33										
34	Free Cash Flow		$6.23	$9.36	$8.96	$9.83	$10.85	$11.78	$12.78	$13.85
35										
36	Terminal Value @ 5%									$184.47
37	Total Free Cash Flow		$6.23	$9.36	$8.96	$9.83	$10.85	$11.78	$12.78	$198.32
38										
39	Discounted Value		$5.62	$7.59	$6.53	$6.42	$6.33	$6.12	$5.90	$81.12
40										
41	Value of Firm	**$125.63**								
42	Non-equity contributions	-59.13								
43	**Value of Equity**	**$66.50**								
44	Implied Debt-to-Value	**47.1%**								

Adjusted Present Value Approach: Value of equity is $57.27 ml vs. $66.5 ml in WACC

Estimate of Value of												
The Unlevered Firm		At Closing	1991	1992	1993	1994	1995	1996	1997	1998		
Free Cash Flow			$6.23	$9.36	$8.96	$9.83	$10.85	$11.78	$12.78	$13.85		
Terminal Value @ 5%										$168.92		
			$6.23	$9.36	$8.96	$9.83	$10.85	$11.78	$12.78	$182.77		
Value of Unlevered	13.61%	$107.04									1999	
senior term	32.00	0.09	1.44	2.70	2.34	1.98	1.58	1.08	0.50			
repayment			2.00	4.00	4.00	4.50	5.50	6.50	5.50			
balance senior	32.00		30.00	26.00	22.00	17.50	12.00	5.50	-			
senior revolver	1.70	0.09	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16		
mezzanine	15.00	0.10	0.75	1.50	1.50	1.50	1.50	1.50	1.50	1.50		
exist liabs			1.00	0.03	0.03	0.03	0.03	0.03	0.03	0.04		
vendor note	11.00	0.08	0.46	0.93	0.93	0.93	0.93	0.93	0.93	0.93		
total interest			$3.82	$5.32	$4.96	$4.60	$4.20	$3.70	$3.12	$2.63		cost of deb
sum of debt by year	59.70		57.70	53.70	49.70	45.20	39.70	33.20	27.70	27.70	14.87	9.00%
Estimate of the Present Value of Debt Tax Shields									net debt	at 98 end	$14.87	
Interest Expense			$3.82	$5.32	$4.96	$4.60	$4.20	$3.70	$3.12	$2.63	$1.34	
Annual Tax Reduction			$1.34	$1.86	$1.74	$1.61	$1.47	$1.30	$1.09	$0.92	$0.47	
Terminal Value of Tax Shields									pvts TV 98	$5.71		
									growing at 5%			
PV of Tax Shields (r_d one year, r_a rest)		$9.35										
										$174.63		
Valuation Summary												
Value of Unlevered Firm		$107.04										
PV Debt Tax Shields		9.35										
Total Value of Firm		116.40										
Non-equity contributions		-59.13										
Value of Equity		$57.27										
		0.50801007										

Gearing ratio

Correct solution: calculate WACC each year starting with last year solving simultaneously for D/V & V

WACC Method Using Miles-Ezzel	At Closing 1990	1991	1992	1993	1994	1995	1996	1997	1998
Estimate of Annual Discount Factors									
Actual Yearly Debt Levels		59.13	55.4	51.87	47.37	41.87	35.37	29.87	14.87
value by year		116.44	124.08	129.77	136.74	143.93	151.27	158.89	166.73
D/V Ratios		50.8%	44.6%	40.0%	34.6%	29.1%	23.4%	18.8%	8.9%
cost of debt by year		9.15%	9.15%	9.16%	9.18%	9.20%	9.22%	9.27%	9.32%
Opportunity cost of Capital	13.61%								
Tax Rate	35%								
Adjusted Cost of Capital		11.92%	12.12%	12.28%	12.45%	12.64%	12.83%	12.98%	13.31%
Annual Discount Factor		0.89	0.80	0.71	0.63	0.56	0.50	0.44	0.39
Free Cash Flow		$6.23	$9.36	$8.96	$9.83	$10.85	$11.78	$12.78	$13.85
			50.2%	-4.3%	9.8%	10.4%	8.5%	8.5%	8.4%
Terminal Value at end of 1998									$175.07
Total Free Cash Flow		$6.23	$9.36	$8.96	$9.83	$10.85	$11.78	$12.78	$188.92
Firm Values at end of Each Year									
discount factor to 90		0.89	0.80	0.71	0.63	0.56	0.50	0.44	0.39
Firm Value at end of 1990	**116.44**	5.57	7.46	6.36	6.21	6.08	5.85	5.62	73.30
discount factor to 91			0.89	0.79	0.71	0.63	0.56	0.49	0.43
Firm Value at end of 1991	**124.08**		8.35	7.12	6.95	6.81	6.55	6.29	82.03
discount factor to 92				0.89	0.79	0.70	0.62	0.55	0.49
Firm Value at end of 1992	**129.77**		-	7.98	7.79	7.63	7.34	7.05	91.98
discount factor to 93					0.89	0.79	0.70	0.62	0.55
Firm Value at end of 1993	**136.74**		-	-	8.74	8.57	8.24	7.92	103.27
discount factor to 94						0.89	0.79	0.70	0.61
Firm Value at end of 1994	**143.93**		-	-	-	9.63	9.27	8.90	116.13
discount factor to 95							0.89	0.78	0.69
Firm Value at end of 1995	**151.27**		-	-	-	-	10.44	10.03	130.80
discount factor to 96								0.89	0.78
Firm Value at end of 1996	**158.89**		-	-	-	-	-	11.31	147.58
discount factor to 97									0.88
Firm Value at end of 1997	**166.73**		-	-	-	-	-	-	166.73
Value of Firm	**$116.44**								
Non-equity contributions	-59.13								
Value of Equity	**$57.31**								

Appendix: Bank of England: Model-based estimates of the historical market return (pre-gold standard vs post gold standard 1931 approx.):

	Gold standard up to 1931		Post-GS	
	Data	**Model**	Data	**Model**
Real rate	3.0 , 3.4	**3.0**	1.1 , 1.7	**1.8**
Equity premium	0.4 , 0.6	**0.3**	7.6 , 8.6	**6.7**
Cons growth	0.8	**1.2**	2.0 , 2.3	**2.2**
Cons vol	1.5 , 1.8	**1.5**	2.3 , 3.1	**5.5**
Cons Skew	-0.2 , 0.2	**-0.0**	-2.1 , -1.3	**-1.2**
Cons Kurtosis	2.3 , 2.4	**2.5**	8.0 , 13.9	**9.4**